TUNAR AHMADZADA

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SUMMARY

11th-grade student pursuing independent work at the intersection of artificial intelligence and computational mathematics. Selected as one of four students nationwide to represent Azerbaijan at the International Olympiad in Artificial Intelligence (IOAI), earning a Gold Medal at the National Round. Research in algorithmic number theory received national recognition at ISEF, while an applied computational project received NASA's Galactic Problem Solver award. Seeking to build on this research-driven foundation through rigorous undergraduate study in Computer Science and Artificial Intelligence.

EDUCATION

High School No. 85 *Sep 2016 – Jun 2027 (expected)*

Baku, Azerbaijan

- GPA: 4.9 / 5
- Ranked 1st in school on the National Middle School Graduation Exam, scoring 292 / 300.

HONORS & AWARDS

- International Olympiad in Artificial Intelligence (IOAI) — National Round, **Gold Medal** (Ranked #3 of 2,800+) 2026
- International Science and Engineering Fair (ISEF) — National Round, **Bronze Medal** (1,500+ participants) 2026
- International AI Challenge for Children and Youth — Main Stage, **Finalist (ongoing)** (10,000+ participants) 2026
- Russian Olympiad in Artificial Intelligence (ROAI) — Main Stage, **Finalist (ongoing)** (~70,000 participants) 2026
- International Economics Olympiad (IEO) — National Round, **Bronze Medal** 2026
- International Philosophy Olympiad (IPO) — National Round, **Silver Medal** 2026
- NASA Space Apps Challenge — **Galactic Problem Solver** (114,000+ participants) 2025
- Others: National Science Olympiads — Finalist: Astronomy, Statistics (2025–2026); Semi-Finalist: Mathematics, Computer Science, Physics, Chemistry (2023–2026) (73,000+ participants)

SKILLS

Python Programming (NumPy, pandas, scikit-learn) • Machine Learning • Arduino & Robotics • Mathematics • Event Coordination • Community Engagement • Volunteer Leadership

EXPERIENCE

Member — The Junior Academy, The New York Academy of Sciences *Sep 2026 – Present*

- Selected for an application-based program that places high school students from around the world into international teams guided by STEM mentors, with the aim of preparing the next generation of innovators.
- Collaborate across countries in 10-week virtual Innovation Challenges: researching, designing, testing, and pitching solutions to real-world problems; currently developing NeuroQuit (see Projects).

Delegate & Participant — International Olympiad in Artificial Intelligence (IOAI) *Aug 2026*

Astana, Kazakhstan

- Represented Azerbaijan's National Team after earning a Gold Medal at the National Round; selected as 1 of 4 students nationwide to advance to the international round.
- Competed at the third IOAI (2–8 Aug 2026), which brought together 466 students from 108 countries and territories, making it the second-largest international science olympiad after the IMO and one centered on hands-on AI problem solving.

Participant — Leva Mathematical Camp (LMK) *Aug 2026 – Present*

Leva Science Community, Türkiye (Remote)

- Admitted to a virtual mathematics camp run by a nonprofit science community after a three-stage selection process; an online cultural and academic exchange with participants from middle school through university level across multiple countries.
- Work on olympiad-style problems and develop mathematical thinking through peer learning with fellow participants.

Local Lead — NASA Space Apps Challenge *July 2026 – Present*

Mingachevir, Azerbaijan

- Lead the Mingachevir Local Event of the NASA International Space Apps Challenge, an annual global hackathon (480,000+ participants across 190+ countries since 2012) where teams use NASA's open data to tackle real challenges on Earth and in space.
- Direct the local chapter's two-day hackathon, coordinating outreach, mentorship, and judging with NASA's Global Organizing Team to connect Azerbaijani innovators with NASA's open data and Space Agency Partners.

Co-Founder & Community Administrator — Texnosfer *June 2026 – Present*

- Co-founded and administer a nonprofit technology community offering free webinars, seminars, and problem-solving discussions to tech enthusiasts of all ages.
- Run free, open-to-all competitions with prizes for community members, with the goal of making technology learning accessible regardless of age or background.

Volunteer Judge — Technovation Girls *Apr 2026 – Jun 2026*

- Evaluated mobile application submissions in Technovation Girls, the world's largest technology-entrepreneurship competition for girls (370,000+ participants across 120+ countries), giving teams scores and feedback in support of its mission to close the gender gap in technology.
- Earned Technovation's Gold Certificate for excellence in evaluation and mentorship.

National Ambassador — International Youth Math Challenge (IYMC) *Mar 2026 – Present*

Baku, Azerbaijan

- Represent IYMC (founded 2018; 30,000+ students from 140+ countries) in Azerbaijan. Every participant solves the same problems from the first round, giving students worldwide a fair international stage.
- Spearheaded local promotion and organized peer study groups, guiding participants through the Qualification and Pre-Final rounds.

PROJECTS

NeuroQuit: An EEG-Based Wearable for Early Nicotine Craving Detection *2026 – Present*

Team project (ongoing) — The Junior Academy, The New York Academy of Sciences

- Developing a non-invasive, ear-worn EEG wearable concept that uses AI to detect early nicotine craving before smoking occurs, addressing the gap that most cessation methods intervene only after a person has decided to smoke.
- Contribute to research and development of the multimodal approach: per-user calibration on baseline and controlled-craving EEG data, machine-learning analysis of alpha and theta activity, and fusion with heart rate and galvanic skin response to reduce false alerts.
- Lead the testing team: design the testing framework with experimental and control groups, run simulation calibration tests with volunteers, and observe practical and result tests.
- Collect volunteer feedback, analyze limitations such as the false-alert rate, and summarize experimental feasibility in data files; evaluation metrics include detection accuracy, precision, recall, and false-alert rate.

Common Divisibility Framework for Prime Numbers *2026*

ISEF National Round — Mathematics Category

- Researched divisibility properties of prime numbers and developed a single generalized algorithm for determining divisibility by any positive integer, unifying classical divisor-specific rules into one method.
- Presented findings at Azerbaijan's national ISEF qualifying round, earning a Bronze Medal in Mathematics against republic-level candidates. Preprint: https://doi.org/10.5281/zenodo.21335786

Space Debris Recycling System — NASA Space Apps Challenge *2025*

Global Hackathon

- Designed a device concept enabling in-orbit space debris to be repurposed as raw material and 3D-printed on-site into spare parts for spacecraft and station maintenance.
- Engineered the computational solution within a 48-hour international hackathon, applying data analysis and rapid-prototyping methods to model the debris-to-part production workflow; awarded NASA's "Galactic Problem Solver" distinction.

CERTIFICATIONS

- Elements of AI — University of Helsinki
- Machine Learning Introduction for Everyone — IBM (Coursera)
- Programming for Everybody — University of Michigan (Coursera)
- Introduction to Microcontrollers with Arduino — METU (ODTÜ)

LANGUAGES

Azerbaijani (Native) • Turkish (C1) • English (C1; TOEFL iBT 100 / 120)

INTERESTS

Mathematics • Robotics • Programming • Research • Reading • Competitions • Theatre • Music (Violin, Piano) • Plant Growing & Gardening • Cooking